

LECTURE 6

5. GRAPH THEORY

5.1. **Introduction.** Graphs are key in the sciences and beyond. Some applications include:

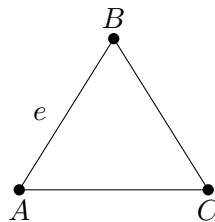
- Social networks used since the 1930's in sociology. A well-known example is Facebook, where vertices = people, edges = friendship links.
- The internet is a huge (directed) graph with vertices = webpages and edges = hyperlinks.
- The Travelling Salesman Problem (TSP): Find a shortest route that visits all cities, where vertices = cities, edges = roads, and the edges are labelled by a distance. The TSP has many modifications. The labels can be time to travel, cost, etc.
- In DNA-sequencing the vertices are DNA-fragments and labels describe the degree of similarity between two fragments.
- Structural rigidity, etc.

Simply put, a graph is a set of vertices, with edges between some of the given vertices. An edge can also have a direction. This respectively gives rise to the notion of a directed graph (a graph where all edges in the graph are directed). Formally:

Definition 5.1. Let V be a finite non-empty set, and let $E \subseteq V \times V$. The pair (V, E) is then called a *directed graph* (on V), where V is the set of *vertices* (or *nodes*), and E is the set of (*directed*) *edges* (or *arcs*).

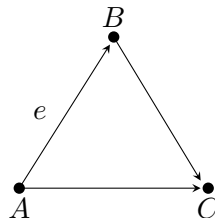
When there is no concern about the direction of any edge, we will still write $G = (V, E)$, but now E is a set of unordered pairs of elements from V , and G is called an *undirected graph* (or simply, a *graph*).

Example 5.2. An example of a graph is:



The edge e can also be referred to as the subset $\{A, B\}$.

An example of a directed graph is:



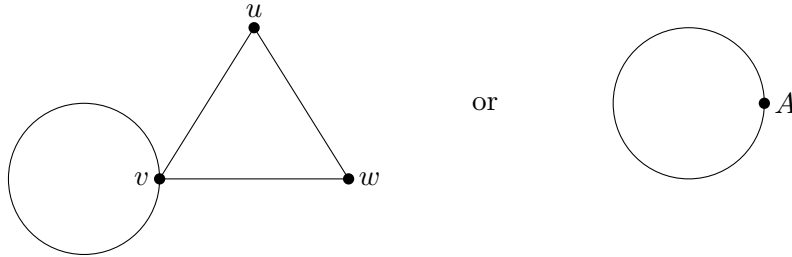
Here the directed edge e can also be referred to as the ordered pair (A, B) .

For a directed or undirected graph $G = (V, E)$, we refer to V as the *vertex set* of G and to E as the *edge set* of G .

Definition 5.3. Let $G = (V, E)$ be a graph. An edge $e \in E$ in G from a vertex v to itself is called a *loop*.

When a graph G contains no loops, it is called *loop-free*.

Example 5.4. An example of a graph with a loop is:



Both graphs in Example 5.2 are loop-free.

Definition 5.5. Let x, y be (not necessarily distinct) vertices in an undirected graph $G = (V, E)$. An $x - y$ *walk* in G is a (loop-free) finite alternating sequence

$$x = x_0, e_1, x_1, e_2, x_2, e_3, \dots, e_{n-1}, x_{n-1}, e_n, x_n = y$$

of vertices and edges from G , starting at the vertex x and ending at the vertex y and involving the n edges $e_i = \{x_{i-1}, x_i\}$, where $1 \leq i \leq n$.

The *length* of this walk is n , which is the number of edges in the walk. (When $n = 0$, we have $x = y$ and there are no edges and the walk is called *trivial*.)

An $x - y$ walk where $x = y$ and $n > 1$ is called a *closed walk*. Otherwise the walk is called *open*.

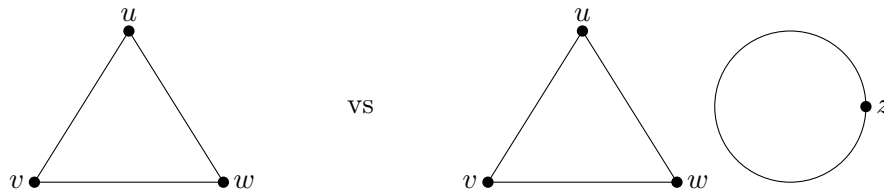
Definition 5.6. For an $x - y$ walk in an undirected graph $G = (V, E)$, if no edge is repeated, then the walk is called an $x - y$ *trail*. A closed $x - x$ trail is called a *circuit*. If no vertex of the $x - y$ walk occurs more than once, then the walk is called an $x - y$ *path*. When $x = y$, the term *cycle* is used to describe such a closed path.

For a directed graph, one uses the terms *directed walks*, *directed paths*, *directed cycles*, etc.

Definition 5.7. Let $G = (V, E)$ be an undirected graph. We call G *connected* if there is a path between any two distinct vertices of G .

A graph that is not connected is called *disconnected*.

Example 5.8. An example of a connected graph and a disconnected graph are respectively:



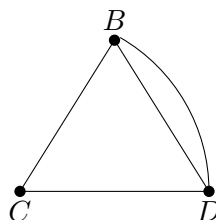
If a graph is disconnected, it is a union of connected subgraphs. These connected subgraphs are called *components*.

Definition 5.9. For any graph $G = (V, E)$, the number of components of G is denoted by $\kappa(G)$.

For the disconnected graph G in Example 5.8, we have $\kappa(G) = 2$.

Definition 5.10. Let V be a finite non-empty set. We say that the pair (V, E) determines a *multigraph* G with vertex set V and edge set $E^\dagger$ if, for some $x, y \in V$, there are two or more edges in E of the form (a) (x, y) (for a directed multigraph), or (b) $\{x, y\}$ (for an undirected multigraph). In either case, we still write $G = (V, E)$ to denote the multigraph.

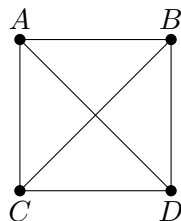
Example 5.11. An example of a multigraph is:



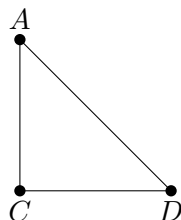
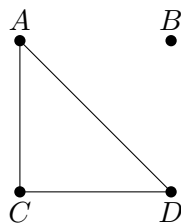
5.2. Subgraphs.

Definition 5.12. If $G = (V, E)$ is a graph (directed or undirected). Then $G_1 = (V_1, E_1)$ is a *subgraph* of G if $V_1 \subseteq V$, $E_1 \subseteq E$ and both endpoints of any edge in E_1 are in V_1 . If $V_1 = V$ it is called a *spanning subgraph*.

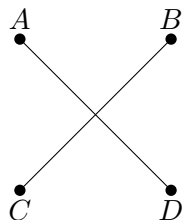
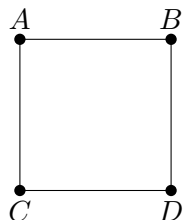
Example 5.13. For the graph



two examples of subgraphs are:

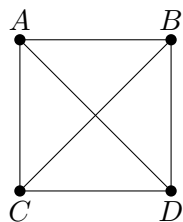


Two examples of spanning subgraphs are:

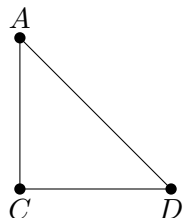


Definition 5.14. Let $G = (V, E)$ be a graph (directed or undirected). If $U \subseteq V$ then $\langle U \rangle$, the *subgraph induced by U* , is $G_1 = (U, E_1)$ where E_1 contains all edges of E that have both endpoints in U .

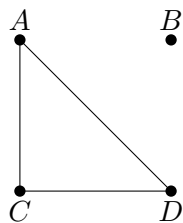
Example 5.15. For the graph



an example of an induced subgraph would be:



But the following is not an induced subgraph:

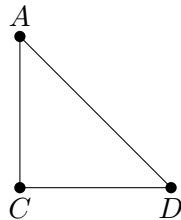


Definition 5.16. For a graph $G = (V, E)$ and $v \in V$, we denote by $G - v$ the induced subgraph from $V - \{v\}$. If $e \in E$ then $G - e$ denotes the subgraph $(V, E - \{e\})$.

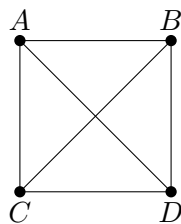
Informally $G - v$ is G with the vertex v (and its edges) removed and $G - e$ is G with the edge e removed.

We denote by K_n the complete graph on n vertices. For K_n all $\binom{n}{2}$ possible edges are in E .

Example 5.17. The graph K_3 is:

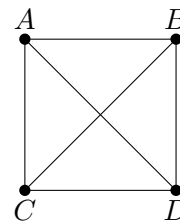
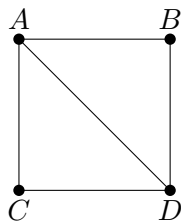


The graph K_4 is:

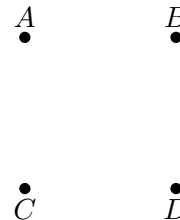
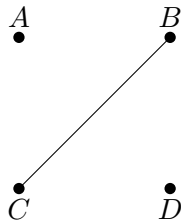


Definition 5.18. For a graph G , we denote the *complement* of G by $\overline{G}$, which is obtained from G by keeping the same set of vertices but putting edges exactly between those (distinct) vertices that were not connected in G .

Example 5.19. For the following two graphs



the complements are respectively:

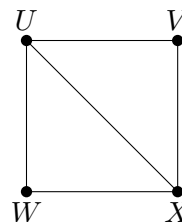
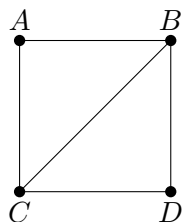


5.3. Isomorphic graphs.

Definition 5.20. Two (undirected) graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be *isomorphic* if there is a bijective function $f : V_1 \rightarrow V_2$ such that $\{a, b\} \in E_1$ if and only if $\{f(a), f(b)\} \in E_2$.

Such a bijective function is called an (*graph*) *isomorphism*.

Example 5.21. The following two graphs are isomorphic

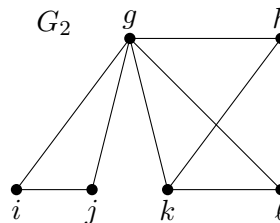
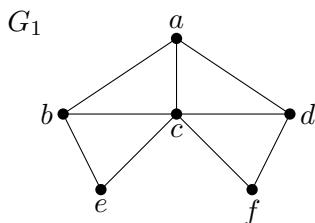


by the isomorphism that sends $A \mapsto W$, $B \mapsto U$, $C \mapsto X$, and $D \mapsto V$, for example.

Isomorphic graphs are structurally the same, subject to moving positions of the vertices you can obtain G_2 from G_1 if $G_1 \cong G_2$.

In particular if $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic then $|V_1| = |V_2|$, $|E_1| = |E_2|$. Also v and $f(v)$ have the same degree (= number of edges incident to a vertex, a loop counts as two). It also follows from $G_1 \cong G_2$ that cycles in G_1 correspond to cycles in G_2 , walks correspond to walks, etc.

Example 5.22. Consider the following graphs:



Are G_1 and G_2 isomorphic?

Solution: Let us denote $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$. We first check if the number of vertices and edges are equal: we have

$$|V_1| = |V_2| = 6 \quad \text{and} \quad |E_1| = |E_2| = 9.$$

Let us then look at the degrees of the vertices. Both G_1 and G_2 have two vertices of degree 2, three vertices of degree 3, and one vertex of degree 5. Thus, so far nothing differs so perhaps they are isomorphic?

However, in G_1 the vertices of degree two are not connected by an edge, whereas in G_2 they are. This means that $G_1 \not\cong G_2$. [Recall that any isomorphism f would have $\{\alpha(e), \alpha(f)\} = \{i, j\}$ and therefore $i \sim j$ but the fact that $e \not\sim f$ yields a contradiction.]

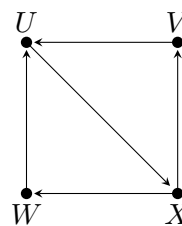
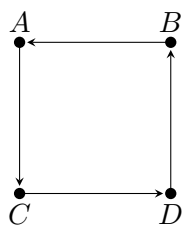
5.4. Euler trails and circuits.

Theorem 5.23. Let $G = (V, E)$ be an undirected graph. If $a, b \in V$ are distinct vertices and there is a trail from a to b , then there is a path from a to b .

Proof. Pick a trail $a \rightarrow b$ of minimal length. If a vertex c is repeated on the trail we can remove the circuit $c \rightarrow c$ on the trail and get a shorter trail. This contradicts that our trail had minimal length so no vertex can be repeated. Then, by definition, our trail is a path. $\square$

Definition 5.24. An *Euler circuit* of an undirected graph (or multigraph) is a closed trail (= circuit) that traverses every edge exactly once. (An open such walk is called an *Euler trail*.)

Example 5.25. The left graph depicts a circuit, whereas the right one a trail (starting at X and ending at U):



We will find out which graphs have an Euler circuit, but let us first look at a basic fact that holds in any graph or multigraph.

Theorem 5.26. If $G = (V, E)$ is a (multi)graph then

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof. Both sides count the total number of endpoints of edges in the graph. $\square$

A direct consequence is that every graph has an even number of vertices with odd degree. Now for the main result.

Theorem 5.27. Let $G = (V, E)$ be an undirected (multi)graph with no isolated vertices. Then G has an Euler circuit if and only if G is connected and every vertex of G is of even degree.

Proof. ($\Rightarrow$) The Euler circuit connects any two vertices so G is connected. By following the circuit and subtracting two from the degree every time we pass a node (or run through a loop at this node) we will end up with no edges left and hence the degree is zero at all nodes. Since we got zero after subtracting 2 a number of times, we started with an even number. (The starting point of the circuit gets one subtraction following the first edge but also one from the final edge of the circuit.) This proves the forward direction.

($\Leftarrow$) We now assume that G is connected and all degrees are even. If $|E| = 1, 2$ we can only get

- one vertex with one loop,
- one vertex with two loops,
- two vertices with two edges between them.

These all have Euler circuits.

We use induction to prove the general case of n edges, so we now assume the statement holds for graphs with less than n edges.

If $|E| = n$ we start at any vertex v and take a longest trail starting at v . Whenever you reach a vertex that is not v you can continue since there is always an odd number of edges left once you have arrived at that vertex. This means that you can only stop at v , so we have found a circuit containing v . Now remove all edges of this circuit and the vertices that become isolated afterwards. By the induction hypothesis each component of the new graph has an Euler circuit. Follow the circuit from v and add the circuits of the corresponding component when you come to a vertex that is in that component. This is an Euler circuit for the original graph. $\square$

Example 5.28. In K_n each vertex has degree $n - 1$ so K_n has an Euler circuit if and only if n is odd.

Corollary 5.29. *If G is an undirected (multi)graph with no isolated vertices then G has an Euler trail if and only if exactly two vertices are of odd degree and G is connected.*

Proof. ($\Leftarrow$) If $a, b \in V$ are of odd degree, add an edge $a \sim b$. By Theorem 5.27, this graph has an Euler circuit which gives us an Euler trail from a to b .

($\Rightarrow$) Assume there is an Euler trail, say from c to d . Connect c and d . Now we have an Euler circuit, so all degrees are even by the previous result. Now all degrees were even from the start except for $\deg(c)$ and $\deg(d)$ which were odd. The circuit also shows that c and d were connected even before we added an edge since we have at least two trails from c to d using the circuit. Hence the original graph was also connected. $\square$